The Landau Quantization Problem using Feynman's Path Integrals

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Contents

1 Feynman's Idea of Path Integrals

Richard Feynman got inspired from Paul Dirac's statement that the transition amplitude to go from one event, say (x,t) to another infinetesimally close event, $(x', t+\epsilon)$ is something like $e^{\frac{i}{\hbar}S}$, where S is the classical action for the particle to go from (x,t) to $(x', t+\epsilon)$. In the light of this remark, Feynman proposed that the probability amplitude for a particle to reach at (x_b, t_b) from (x_a, t_a) is the sum of individual amplitudes contributed by each and every possible paths from (x_a, t_a) to (x_b, t_b), and all the paths contribute to the total amplitude in equal magnitudes but in different phases. The phase is proportional to the classical action of that particular trajectory. Mathematically,

$$K(x_b, t_b; x_a, t_a) = \sum_{all\, paths\, from\, a\, to\, b} \phi[x(t)] \tag{1.1}$$

where,

$$\phi[x(t)] \propto e^{\frac{i}{\hbar}S[x(t)]} \tag{1.2}$$

$K(x_b, t_b; x_a, t_a)$ is the total amplitude. It is called the *Propagator* or *Kernal*. We denote it by $K(b,a)$ for simplicity.

1.1 The Classical Limit

Even before constructing the path integral rigorousely, the classical limit is evident. The classical path of a particle is one particular path that minimizes the action. i.e, any first order change in the trajectory does not change the action to first order.

Now consider any path other than the classical path. For a classical system, $S >> \hbar$ and hence the phase angle is huge. Any small change in the trajectory will change the action and hence the phase angle changes very rapidly which in turn will make the individual amplitudes oscillate from -ve to +ve rapidly . So, if any path contribute +ve to the total amplitude, another infinitesimally close path will contribute equally -ve and the net effect cancels. But for the classical path, small changes in the path do not change the action upto first order. So paths infinitesimally close to the classical path contribute in nearly equal phases, $e^{iS_{cl}/\hbar}$. So the contributions from other paths cancels out and only the classical path contribute to the total amplitude for a classical system.

1.2 Sum Over Paths

Divide the independent variable time into N intervals of length ϵ. $\epsilon = \frac{t_b - t_a}{N} = t_{i+1} - t_i$. Choose, at each t_i an x_i. Connect all these x_i to get a path. This is a path that contributes to the propagator. Now vary all x_i except the end points and take the sum. This sum is just over one particular subset of all paths. Take $N \to \infty (\epsilon \to 0)$ to get the whole paths. A normalization constant A is introduced for each integral. So we get the final result as,

$$K(b,a) = \lim_{N\to\infty} A^{-N} \iint \cdots \int exp\left\{\frac{i}{\hbar}S[b,a]\right\} dx_1 dx_2 \cdots dx_{N-1} = \int_a^b exp\left\{\frac{i}{\hbar}S[b,a]\right\} \mathcal{D}x(t) \tag{1.3}$$

where,

$$\int_a^b \mathcal{D}x(t) \equiv \lim_{N\to\infty} A^{-N} \iint \cdots \int dx_1 dx_2 \cdots dx_{N-1}$$

1.3 Events In Succession

Choose any (x_c, t_c) between (x_a, t_a) and (x_b, t_b). As action is a time integral, we can break the integral into two parts, one from t_a to t_c and the other from t_c to t_b. Therefore,

$$S[b,a] = S[b,c] + S[c,a] \tag{1.4}$$

$$\Rightarrow K(b,a) = \int_a^b exp\left\{\frac{i}{\hbar}S[b,c] + \frac{i}{\hbar}S[c,a]\right\}\mathcal{D}x(t) \tag{1.5}$$

The path can be splitted in two, one from a to c and the other from c to b. We first carry out the integral from a to c and integrate over all possible x_c. While doing this, $e^{\frac{i}{\hbar}S[b,c]}$ is constant. Hence,

$$K(b,a) = \int_{-\infty}^{\infty}\int_c^b exp\left\{\frac{i}{\hbar}S[b,c]\right\}K(c,a)\mathcal{D}x(t)\,dx_c \tag{1.6}$$

Now we carry out the integral from c to b to get,

$$K(b,a) = \int_{-\infty}^{\infty} K(b,c)K(c,a)\,dx_c \tag{1.7}$$

This is the rule for two events in succession. It can be extented to more than two events. i.e,

$$K(b,a) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} K(b,d)K(d,c)K(c,a)\,dx_c\,dx_d \tag{1.8}$$

2 Equivalence With Other Approaches

The method of path integrals developed by Feynman is equivalent to other known approaches of quantum mechanics.

2.1 The Wave Function

The wave function can be thought as the propagator to reach a particular position from an unspecified starting point. We denote it by $\psi(x,t)$.As the wavefunction is also a propagator, it must obey the rules of propagators. If an initial state $\psi(x_a,t_a)$ is given, then the amplitude to reach (x_b,t_b) can be found using equation 1.7 as,

$$\psi(x_b,t_b) = \int_{-\infty}^{\infty} K(x_b,t_b;x_a,t_a)\psi(x_a,t_a)dx_a \tag{2.1}$$

2.2 The Schrodinger Equation

Equation 2.1 is valid between any two time intervals, in particular between two infinitesimally close times say t and $t+\epsilon$. We consider a system whose action has the property specified by equation 1.4. Let $\psi(y,t)$ and $\psi(x,t+\epsilon)$ be the initial and final states respectively. As ϵ is very small, the velocity and position of the particle can be approximated by $\frac{x-y}{\epsilon}$ and $\frac{x+y}{2}$ respectively. The action in that small time interval is,

$$S \simeq \epsilon\mathcal{L}\left(\frac{x-y}{\epsilon}, \frac{x+y}{2}\right)$$

Therefore,

$$K(x,t+\epsilon;y,t) = \frac{1}{A}exp\left\{\frac{i}{\hbar}\epsilon\mathcal{L}\left(\frac{x-y}{\epsilon}, \frac{x+y}{2}\right)\right\}$$

Hence by equation 2.1, we get,

$$\psi(x,t+\epsilon) = \int_{-\infty}^{\infty}\frac{1}{A}exp\left\{\frac{i}{\hbar}\epsilon\mathcal{L}\left(\frac{x-y}{\epsilon}, \frac{x+y}{2}\right)\right\}\psi(y,t)dy \tag{2.2}$$

Take $\mathcal{L}(\dot{x},x,t) = \frac{m}{2}\dot{x}^2 - V(x,t)$

$$\therefore S \simeq \epsilon\left[\frac{m}{2}\left(\frac{x-y}{\epsilon}\right)^2 - V\left(\frac{x+y}{2},t\right)\right] = \frac{m(x-y)^2}{2\epsilon} - \epsilon V\left(\frac{x+y}{2},t\right) \tag{2.3}$$

$$\Rightarrow \psi(x,t+\epsilon) = \frac{1}{A}\int_{-\infty}^{\infty} exp\left\{\frac{i}{\hbar}\frac{m(x-y)^2}{2\epsilon}\right\} exp\left\{-\frac{i}{\hbar}\epsilon V\left(\frac{x+y}{2},t\right)\right\}\psi(y,t)dy \tag{2.4}$$

Substitute $y = x+\eta$. Therefore the integral changes as,

$$\psi(x,t+\epsilon) = \frac{1}{A}\int_{-\infty}^{\infty} exp\left\{\frac{im\eta^2}{2\hbar\epsilon}\right\} exp\left\{-\frac{i}{\hbar}\epsilon V\left(x+\frac{\eta}{2},t\right)\right\}\psi(x+\eta,t)d\eta \tag{2.5}$$

The integral will give significant value only when η^2 is of the order of ϵ. Otherwise $\frac{\eta^2}{\epsilon}$ is huge and hence the exponent will oscillate rapidly between +ve and -ve hence cancels out the net result for $\eta >> \epsilon$. As the exponent is of the form $\frac{\eta^2}{\epsilon}$, first order change in ϵ should be compensated by second order change in η. We expand things upto first order in ϵ and second order in η.

$$\epsilon V\left(x+\frac{\eta}{2},t\right) = \epsilon V(x,t) + \frac{\eta\epsilon}{2}\frac{\partial V}{\partial x} \simeq \epsilon V(x,t)$$

$$\psi(x,t) + \epsilon\frac{\partial\psi(x,t)}{\partial t} = \frac{1}{A}\int_{-\infty}^{\infty} exp\left\{\frac{im\eta^2}{2\hbar\epsilon}\right\}\left(1-\frac{i}{\hbar}\epsilon V(x,t)\right)\left(\psi(x,t)+\eta\frac{\partial\psi(x,t)}{\partial x}+\frac{\eta^2}{2}\frac{\partial^2\psi(x,t)}{\partial x^2}\right)d\eta \tag{2.6}$$

Using the integrals,

$$\int_{-\infty}^{\infty} e^{ax^2}dx = \sqrt{\frac{\pi}{-a}} \quad ; \int_{-\infty}^{\infty} xe^{ax^2}dx = 0 \quad ; \int_{-\infty}^{\infty} x^2e^{ax^2}dx = \frac{-i\sqrt{\pi}}{2a\sqrt{a}} \tag{2.7}$$

We get,

$$\psi + \epsilon\frac{\partial\psi}{\partial t} = \frac{1}{A}\left[\psi\sqrt{\frac{2\pi i\hbar\epsilon}{m}} - \frac{i}{\hbar}\epsilon V\psi\sqrt{\frac{2\pi i\hbar\epsilon}{m}} + \frac{1}{2}\left(-\frac{\hbar\epsilon}{m}\sqrt{\frac{2\pi\hbar\epsilon}{im}}\right)\frac{\partial^2\psi}{\partial x^2} + \frac{i\epsilon^2}{2m}V\sqrt{\frac{2\pi\hbar\epsilon}{im}}\frac{\partial^2\psi}{\partial x^2}\right] \tag{2.8}$$

The last term in RHS is neglected as it is of higher order. Equating coefficients of ψ on both sides yields,

$$A = \sqrt{\frac{2\pi i\hbar\epsilon}{m}} \tag{2.9}$$

Further simplifiaction gives,

$$\psi + \epsilon\frac{\partial\psi}{\partial t} = \psi - \frac{i\epsilon}{\hbar}V\psi + \frac{i\hbar\epsilon}{2m}\frac{\partial^2\psi}{\partial x^2} \tag{2.10}$$

Rearranging terms,

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi \tag{2.11}$$

Which is the Schrodinger Equation.
Hence, path integral approach is equivalent to other approaches.

2.3 Propagator in terms of energy eigen values

The propagator can be represented using energy eigen values. Let $|\psi,t_a\rangle$ and $|\psi,t_b\rangle$ be the initial and final states, and $\mathcal{U}(t_b,t_a)$ be the time evolution operator. Let A be an operator which has simultaneous eigen kets with the energy operator H, denote them by $|a_n\rangle$.

$$|\psi,t_b\rangle = \mathcal{U}(t_b,t_a)|\psi,t_a\rangle = e^{-\frac{i}{\hbar}H(t_b-t_a)}|\psi,t_a\rangle \tag{2.12}$$

We use the completeness relation, $\sum_n |a_n\rangle\langle a_n| = 1$ and $H|a_n\rangle = E_n|a_n\rangle$ to get,

$$|\psi,t_b\rangle = \sum_n |a_n\rangle\langle a_n|\psi,t_a\rangle e^{-\frac{i}{\hbar}E_n(t_b-t_a)} \tag{2.13}$$

We now find the position representation as,

$$\langle x_b|\psi,t_b\rangle = \sum_n \langle x_b|a_n\rangle\langle a_n|\psi,t_a\rangle e^{-\frac{i}{\hbar}E_n(t_b-t_a)} \tag{2.14}$$

Now use,

$$\int_{-\infty}^{\infty} |x_a\rangle \langle x_a| \, dx_a = 1$$

$$\therefore \langle a_n|\psi, t_a\rangle = \int_{-\infty}^{\infty} \langle a_n|x_a\rangle \langle x_a|\psi, t_a\rangle \, dx_a \tag{2.15}$$

Let $\langle x|a_n\rangle = \phi_n(x)$ be the eigen functions of A and denote $\langle x|\psi, t\rangle$ by $\psi(x,t)$. Hence we get,

$$\psi(x_b, t_b) = \int_{-\infty}^{\infty} dx_a \left(\sum_n \phi_n(x_b)\phi_n^*(x_a) e^{-\frac{i}{\hbar}E_n(t_b - t_a)} \right) \psi(x_a, t_a) \tag{2.16}$$

This equation resembles equation 2.1. Hence we get,

$$K(x_b, t_b; x_a, t_a) = \sum_n \phi_n(x_b)\phi_n^*(x_a) e^{-\frac{i}{\hbar}E_n(t_b - t_a)} \tag{2.17}$$

3 The Free Particle

We use a very direct method to compute the propagator of the free particle. The lagrangian is given by,

$$\mathcal{L} = \frac{m}{2}\dot{x}^2 \tag{3.1}$$

We divide time into slices of width ϵ as done before. Hence $\dot{x}_i = \frac{x_i - x_{i-1}}{\epsilon}$. Therefore,

$$\int_{t_a}^{t_b} \frac{m}{2}\dot{x}^2 dt \to \sum_{i=1}^{N} \frac{m}{2}\left(\frac{x_i - x_{i-1}}{\epsilon}\right)^2 \epsilon = \frac{m}{2\epsilon}\sum_{i=1}^{N}(x_i - x_{i-1})^2 \tag{3.2}$$

We have already found the normalizing constant as $A = \sqrt{\frac{2\pi i\hbar\epsilon}{m}}$. Therefore we get,

$$K(b,a) = \lim_{\epsilon\to 0}\left(\frac{m}{2\pi i\hbar\epsilon}\right)^{N/2} \int \cdots \iint exp\left\{\frac{im}{2\hbar\epsilon}\sum_{i=1}^{N}(x_i - x_{i-1})^2\right\} dx_1 \cdots dx_{N-1} \tag{3.3}$$

We first do the integrals and then take the limit.
First do the integral over x_1 only. i.e,

$$I_1 = \left(\frac{m}{2\pi i\hbar\epsilon}\right)^{2/2} \int_{-\infty}^{\infty} e^{\frac{im}{2\hbar\epsilon}[(x_1 - x_0)^2 + (x_2 - x_0)^2]} dx_1 \tag{3.4}$$

Using the result,

$$\int_{-\infty}^{\infty} e^{a(x_1 - x)^2} e^{b(x_2 - x)^2} = \sqrt{\frac{-\pi}{a+b}} e^{\frac{ab}{a+b}(x_1 - x_2)^2} \tag{3.5}$$

We get,

$$I_1 = \left(\frac{m}{2\pi i\hbar \cdot 2\epsilon}\right)^{1/2} e^{\frac{im}{2\hbar\cdot 2\epsilon}(x_2 - x_0)^2} \tag{3.6}$$

Multiply the result with $\left(\frac{m}{2\pi i\hbar\epsilon}\right)^{1/2} e^{\frac{im}{2\hbar\epsilon}(x_3 - x_2)^2}$ and integrate over x_2 in the same manner to get,

$$I_2 = \left(\frac{m}{2\pi i\hbar \cdot 3\epsilon}\right)^{1/2} e^{\frac{im}{2\hbar\cdot 3\epsilon}(x_3 - x_0)^2} \tag{3.7}$$

Proceeding similary $n-1$ steps, We get,

$$I_n = \left(\frac{m}{2\pi i\hbar \cdot n\epsilon}\right)^{1/2} e^{\frac{im}{2\hbar\cdot n\epsilon}(x_n - x_0)^2} \tag{3.8}$$

Now we apply the limit $n \to \infty$. As $n\epsilon = t_b - t_a$ and $x_n - x_0 \to x_b - x_0 = x_b - x_a$ as $n \to \infty$, we get,

$$\lim_{n\to\infty} I_n = \left(\frac{m}{2\pi i\hbar(t_b - t_a)}\right)^{1/2} e^{\frac{im}{2\hbar(t_b - t_a)}(x_b - x_a)^2}$$

$$\therefore K(b,a) = \left(\frac{m}{2\pi i\hbar(t_b - t_a)}\right)^{1/2} exp\left\{\frac{im}{2\hbar(t_b - t_a)}(x_b - x_a)^2\right\} \tag{3.9}$$

4 Gaussian Integrals - A Technique to Evaluate Path Integrals

The direct method to calculate path integrals is a bit difficult. There are easier ways to calculate it. Consider the lagrangian of the form,

$$\mathcal{L} = a(t)\dot{x}^2 + b(t)\dot{x}x + c(t)x^2 + d(t)\dot{x} + e(t)x + f(t) \tag{4.1}$$

We intent to find the propagator for this particular lagrangian. Let $\bar{x}(t)$ be the classical path, denote $S_{cl} = S[\bar{x}(t)]$. Write any path $x(t)$ as,

$$x(t) = \bar{x}(t) + y(t) \tag{4.2}$$

$y(t)$ is the deviation from the classical path. Note that $y(t_b) = y(t_a) = 0$. As x and y differ by a constant $\bar{x}$ at each point, $dx_i = dy_i$ for all intervals of time. Which gives $\mathcal{D}x(t) = \mathcal{D}y(t)$.

$$S[x(t)] = S[\bar{x}(t) + y(t)] \tag{4.3}$$

$$= \int_{t_a}^{t_b} [a(t)(\dot{\bar{x}}^2 + 2\dot{\bar{x}}\dot{y} + \dot{y}^2) + \cdots]dt \tag{4.4}$$

If we collect only the terms which involve only x, it is equal to S_{cl}. If we collect the terms having y upto first order only, the integral will result in zero. It is because the fact that first order change from classical path can only make higher order change in the action. First order change will not be there in the action. It is obvious from the least action principle.

$$\therefore S[x(t)] = S_{cl} + \int_{t_a}^{t_b} [a(t)\dot{y}^2 + b(t)y\dot{y} + c(t)y^2]dt \tag{4.5}$$

$$\Rightarrow K(b,a) = \int_a^b exp\left\{iS/\hbar\right\}\mathcal{D}x = exp\left(\frac{i}{\hbar}S_{cl}\right)\int_0^0 exp\left\{\frac{i}{\hbar}\int_{t_a}^{t_b}[a(t)\dot{y}^2 + b(t)y\dot{y} + c(t)y^2]dt\right\}\mathcal{D}y \tag{4.6}$$

As all paths $y(t)$ starts from $(0, t_a)$ and ends at $(0, t_b)$, it is just a number that depends on the times at end points. Denote it by $F(t_b, t_a)$. Hence we write,

$$K(b,a) = F(t_b, t_a)e^{\frac{i}{\hbar}S_{cl}} \tag{4.7}$$

$F(t_b, t_a)$ is actually equal to the propagator for reaching $(0, t_b)$ from $(0, t_a)$ (or propagator when $x_b = x_a$). Therefore F is function of the difference of times at the end points. Let $T = t_b - t_a$. F can be evaluated using 1.7.

5 The Free Particle Again!

We have,

$$\mathcal{L} = \frac{m}{2}\dot{x}^2 \tag{5.1}$$

The classical trajectory is given by,

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = 0 \tag{5.2}$$

$$\Rightarrow \ddot{x} = 0 \tag{5.3}$$

$$\therefore S_{cl} = \int_{t_a}^{t_b} \frac{m}{2}\dot{x}^2 dt = \frac{m(x_b - x_a)^2}{2(t_b - t_a)} \tag{5.4}$$

$$\therefore K(b,a) = K(0, t_b; 0, t_a)exp\left\{\frac{im(x_b - x_a)^2}{2\hbar(t_b - t_a)}\right\} \tag{5.5}$$

Now to find F, let (x_c, t_c) be an inermediate position. Let $t_b - t_c = t$ and $t_c - t_a = s$. Using equation 1.7,

$$K(0,t_b;0,t_a) = \int_{-\infty}^{\infty} K(0,t_b;x_c,t_c)K(x_c,t_c;0,t_a)dx_c \tag{5.6}$$

$$\Rightarrow F(T) = \int_{-\infty}^{\infty} F(t)exp\left\{\frac{imx_c^2}{2\hbar t}\right\}F(s)exp\left\{\frac{imx_c^2}{2\hbar s}\right\}dx_c \tag{5.7}$$

Using equation 2.7, we get,

$$F(T) = F(t)F(s)i\sqrt{\frac{2\pi\hbar st}{imT}} \tag{5.8}$$

Now choose,

$$F(t) = \sqrt{\frac{m}{2\pi i\hbar t}}f(t) \tag{5.9}$$

Substituting it into the above will yield,

$$f(T) = f(t+s) = f(t)f(s) \tag{5.10}$$

$f(t) = e^{at}$ have the above property. We choose $a = 0$. Hence we get $F(T) = \sqrt{\frac{m}{2\pi i\hbar t}}$. Hence we have got the same propagator as obtained before.

6 The Simple Harmonic Oscillator

6.1 The Propagator

The SHO has the lagrangian given by,

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2x^2 \tag{6.1}$$

Using Euler Lagrange equations, we get the equation of motion for SHO as,

$$\ddot{x} = -\omega^2 x \tag{6.2}$$

It has the general solution,

$$x(t) = A\cos\omega t + B\sin\omega t \tag{6.3}$$

Using the boundary conditions, $x(0) = x_a$ and $x(T) = x_b$, we get,

$$A = x_a \tag{6.4}$$

$$B = \frac{x_b - x_a\cos\omega T}{\sin\omega T} \tag{6.5}$$

$$\therefore x(t) = x_a\cos\omega t + \frac{x_b - x_a\cos\omega T}{\sin\omega T}\sin\omega t \tag{6.6}$$

Now, to find S_{cl}, first do,

$$\frac{m}{2}\int_0^T \dot{x}^2dt = \frac{m}{2}\left[(x\dot{x})_0^T - \int_0^T x\ddot{x}dt\right] \tag{6.7}$$

$$= \frac{m}{2}\left[(x\dot{x})_0^T - \int_0^T x(-\omega^2x)dt\right] \tag{6.8}$$

$$= \frac{m}{2}(x\dot{x})_0^T + \frac{m}{2}\int_0^T \omega^2x^2dt \tag{6.9}$$

$$\therefore S_{cl} = \int_0^T\left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2x^2\right)dt = \frac{m}{2}(x\dot{x})_0^T = \frac{m}{2}\left[x(T)\dot{x}(T) - x(0)\dot{x}(0)\right] \tag{6.10}$$

From the solution for the classical trajectory, we get,

$$\dot{x}(t) = -\omega x_a \sin\omega t + \frac{\omega(x_b - x_a\cos\omega T)}{\sin\omega T}\cos\omega t \tag{6.11}$$

Substituting this to the above and simplifying we get,

$$S_{cl} = \frac{m\omega}{2\sin\omega T}\left[(x_b^2 + x_a^2)\cos\omega T - 2x_a x_b\right] \tag{6.12}$$

So, Using equation 4.7,

$$\therefore K(x_b, T; x_a, 0) = F(T)exp\left\{\frac{im\omega}{2\hbar\sin\omega T}\left[(x_b^2 + x_a^2)\cos\omega T - 2x_a x_b\right]\right\} \tag{6.13}$$

To find the value of $F(T) = K(0, t_b; 0, t_a)$, we use equation 1.7,

$$K(0, t_b; 0, t_a) = \int_{-\infty}^{\infty} K(0, t_b; x_c, t_c)K(x_c, t_c; 0, t_a)dx_c \tag{6.14}$$

Take $t_b - t_c = t$ and $t_c - t_a = s$.

$$\therefore F(T) = \int_{-\infty}^{\infty} F(t)exp\left\{\frac{im\omega}{2\hbar\sin\omega t}\left[x_c^2\cos\omega t\right]\right\} F(s)exp\left\{\frac{im\omega}{2\hbar\sin\omega s}\left[x_c^2\cos\omega s\right]\right\} dx_c \tag{6.15}$$

Using equation 2.7, we get,

$$F(T) = F(t)F(s) \quad i\sqrt{\frac{\pi}{\frac{im\omega}{2\hbar}\left(\frac{\cos\omega t}{\sin\omega t} + \frac{\cos\omega s}{\sin\omega s}\right)}} \tag{6.16}$$

$$= F(t)F(s) \quad i\sqrt{\frac{2\pi\hbar\sin\omega t\sin\omega s}{im\omega\sin\omega T}} \tag{6.17}$$

Take a choice for $F(t)$ as,

$$F(t) = \sqrt{\frac{m\omega}{2\pi i\hbar\sin\omega t}}f(t) \tag{6.18}$$

Substitute this to the equation above, and cancel out the terms to get,

$$f(T) = f(t)f(s) \tag{6.19}$$

As in the case of free particle, we choose $f(t) = 1$ for all t. Hence we get,

$$F(T) = \sqrt{\frac{m\omega}{2\pi i\hbar\sin\omega T}} \tag{6.20}$$

Which in turn gives the propagator as,

$$K(x_b, T; x_a, 0) = \sqrt{\frac{m\omega}{2\pi i\hbar\sin\omega T}}\, exp\left\{\frac{im\omega}{2\hbar\sin\omega T}\left[(x_b^2 + x_a^2)\cos\omega T - 2x_a x_b\right]\right\} \tag{6.21}$$

6.2 Energy Eigen Values For SHO

We expand the propagator using the results,

$$\cos\omega T = \frac{1}{2}e^{i\omega T}(1 + e^{-2i\omega T}) \tag{6.22}$$

$$i\sin\omega T = \frac{1}{2}e^{i\omega T}(1 - e^{-2i\omega T}) \tag{6.23}$$

So using the above in equation 6.21 we get the propagator as,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} \left(1-e^{-2i\omega T}\right)^{-1/2} exp\left\{-\frac{m\omega}{2\hbar}\left[(x_a^2+x_b^2)\frac{1+e^{-2i\omega T}}{1-e^{-2i\omega T}} - \frac{4x_a x_b e^{-i\omega T}}{1-e^{-2i\omega T}}\right]\right\} \tag{6.24}$$

We know that,

$$\frac{1}{1-e^{-2i\omega T}} = 1+e^{-2i\omega T}+\cdots \tag{6.25}$$

and,

$$\frac{1+e^{-2i\omega T}}{1-e^{-2i\omega T}} = 1+2e^{-2i\omega T}+\cdots \tag{6.26}$$

We use these to expand the propagator. We only expand upto $n=2$. As there is an $e^{-i\omega T/2}$ in the beggining of the expansion, the n^{th} term of the expansion will have an $e^{-(n+\frac{1}{2})i\omega T}$. Using equation 2.17, we dirctly get,

$$E_n = (n+\frac{1}{2})\hbar\omega \tag{6.27}$$

Now let us expand the propagator. So we get,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} (1+\frac{1}{2}e^{-2i\omega T}+\cdots) exp\left\{-\frac{m\omega}{2\hbar}\left[(x_a^2+x_b^2)(1+2e^{-2i\omega T}+\cdots) - 4x_a x_b(e^{-i\omega T}+\cdots)\right]\right\} \tag{6.28}$$

Which simplifies to,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} (1+\frac{1}{2}e^{-2i\omega T}+\cdots) exp\left\{-\frac{m\omega}{2\hbar}(x_a^2+x_b^2)\right\} \frac{exp\left\{\frac{2m\omega}{\hbar}x_a x_b e^{-i\omega T}\right\}}{exp\left\{\frac{m\omega(x_a^2+x_b^2)}{\hbar}e^{-2i\omega T}\right\}} \tag{6.29}$$

Expanding the numarator and denominator upto $n=2$ and we get the propagator as,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} (1+\frac{1}{2}e^{-2i\omega T}+\cdots) \times$$
$$exp\left\{-\frac{m\omega}{2\hbar}(x_a^2+x_b^2)\right\}\left[\left(1+\frac{2m\omega}{\hbar}x_a x_b e^{-i\omega T}+\frac{1}{2}\left(\frac{2m\omega}{\hbar}\right)^2 x_a^2 x_b^2 e^{-2i\omega T}\right)\left(1-\frac{m\omega}{\hbar}(x_a^2+x_b^2)e^{-2i\omega T}\right)\right] \tag{6.30}$$

Now we select the lowest term ($n=0$),

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} exp\left\{-\frac{m\omega}{2\hbar}(x_a^2+x_b^2)\right\} = e^{-\frac{i}{\hbar}E_0 T}\phi_0(x_b)\phi_0^*(x_a) \tag{6.31}$$

Take,

$$\phi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} exp\left(-\frac{m\omega x^2}{2\hbar}\right) \tag{6.32}$$

This is the ground state wave function of SHO with energy $E_0 = \frac{1}{2}\hbar\omega$. The next order term ($n=1$) term is,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{\frac{-i\omega T}{2}} e^{-\frac{m\omega}{2\hbar}(x_a^2+x_b^2)} \frac{2m\omega}{\hbar} x_a x_b\, e^{-i\omega T} \tag{6.33}$$

Hence we get,

$$\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{3}{2}i\omega T} \frac{2m\omega}{\hbar} x_a x_b e^{-\frac{m\omega}{2\hbar}(x_a^2+x_b^2)} = e^{-\frac{i}{\hbar}E_1 T}\phi_1(x_b)\phi_1^*(x_a) \tag{6.34}$$

So we get,

$$\phi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(\frac{2m\omega}{\hbar}\right)^{1/2} x\, exp\left(-\frac{m\omega x^2}{2\hbar}\right) = \left(\frac{2m\omega}{\hbar}\right)^{1/2} x\,\phi_0(x) \tag{6.35}$$

It is associated with energy $E_1 = \frac{3}{2}\hbar\omega$.
In this way, all the eigen states can be obained in principle.

7 The Landau Quantization

This problem is the quantization of energy levels of an electron moving in an external magnetic field. Classically the orbits of the electron in an external magnetic field is called cyclotron orbits. But quantum mechanically, the electron cannot have any energy, but the energy levels are quantized.
Consider an electron. Let there be a uniform magnetic field of magnitude B along the z axis. The classical lagrangian of a particle of charge q and moving in a vector potential $\vec{A}$ is given by,

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - \frac{q}{c}\vec{v}\cdot\vec{A} \tag{7.1}$$

We use the gauge $\vec{A} = -\frac{By}{2}\hat{i} + \frac{Bx}{2}\hat{j}$. Hence we get,

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + \frac{eB}{2c}(x\dot{y} - y\dot{x}) \tag{7.2}$$

The equations of motion for the x coordinate is obtained by,

$$\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = 0 \tag{7.3}$$

$$\Rightarrow \frac{eB}{2c}\dot{y} - m\ddot{x} + \frac{eB}{2c}\dot{y} = 0 \tag{7.4}$$

$$\Rightarrow \ddot{x} = \frac{eB}{mc}\dot{y} \tag{7.5}$$

Let $\omega = \frac{eB}{mc}$, the cyclotron frequency. Therefore,

$$\ddot{x} = \omega\dot{y} \tag{7.6}$$

Similarly, the y equations are obtained as,

$$\ddot{y} = -\omega\dot{x} \tag{7.7}$$

and the z equations as,

$$\ddot{z} = 0 \tag{7.8}$$

The z equations can be solved easily. Using the initial conditions $z(0) = z_a$ and $z(T) = z_b$, we get,

$$z(t) = \frac{z_b - z_a}{T}t + z_a \tag{7.9}$$

To solve the coupled equations $\ddot{x} = \omega\dot{y}$ and $\ddot{y} = -\omega\dot{x}$, choose $\dot{x} = v_x$ and $\dot{y} = v_y$. Therefore,

$$\dot{v}_x = \omega v_y \tag{7.10}$$

$$\dot{v}_y = -\omega v_x \tag{7.11}$$

Substituting one in the other, we get,

$$\ddot{v}_x = -\omega^2 v_x \tag{7.12}$$

Which has the solution,

$$v_x(t) = \alpha\cos\omega t + \beta\sin\omega t \tag{7.13}$$

Integrating the above equation one more time, we get,

$$x(t) = A_1\cos\omega t + B_1\sin\omega t + C_1 \tag{7.14}$$

Similarly for the y case we get,

$$y(t) = A_2\cos\omega t + B_2\sin\omega t + C_2 \tag{7.15}$$

Apply these solutions in equation 7.6,

$$\omega^2(-A_1\cos\omega t - B_1\sin\omega t) = \omega^2(-A_2\sin\omega t + B_2\cos\omega t) \tag{7.16}$$

By comparison,

$$A_2 = B_1 \tag{7.17}$$
$$B_2 = -A_1 \tag{7.18}$$

Hence we get the solutions as,

$$x(t) = A_1 \cos\omega t + B_1 \sin\omega t + C_1 \tag{7.19}$$
$$y(t) = B_1 \cos\omega t - A_1 \sin\omega t + C_2 \tag{7.20}$$

Now we apply the initial conditions. We get 4 equations for the four unknowns A_1, B_1, C_1, C_2. It can be arranged in a matrix form as,

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \cos\omega T & \sin\omega T & 1 & 0 \\ -\sin\omega T & \cos\omega T & 0 & 1 \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \\ C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} x_a \\ y_a \\ x_b \\ y_b \end{pmatrix} \tag{7.21}$$

The inverse of the coefficient matrix is obtained as,

$$\frac{1}{2} \begin{pmatrix} 1 & \frac{1}{\tan\frac{\omega T}{2}} & -1 & -\frac{1}{\tan\frac{\omega T}{2}} \\ -\frac{1}{\tan\frac{\omega T}{2}} & 1 & \frac{1}{\tan\frac{\omega T}{2}} & -1 \\ 1 & -\frac{1}{\tan\frac{\omega T}{2}} & 1 & \frac{1}{\tan\frac{\omega T}{2}} \\ \frac{1}{\tan\frac{\omega T}{2}} & 1 & -\frac{1}{\tan\frac{\omega T}{2}} & 1 \end{pmatrix} \tag{7.22}$$

So we get,

$$A_1 = \frac{1}{2}\left(x_a - x_b + \frac{y_a - y_b}{\tan\frac{\omega T}{2}}\right) \tag{7.23}$$
$$B_1 = \frac{1}{2}\left(y_a - y_b + \frac{x_b - x_a}{\tan\frac{\omega T}{2}}\right) \tag{7.24}$$
$$C_1 = \frac{1}{2}\left(x_a + x_b + \frac{y_b - y_a}{\tan\frac{\omega T}{2}}\right) \tag{7.25}$$
$$C_2 = \frac{1}{2}\left(y_a + y_b + \frac{x_a - x_b}{\tan\frac{\omega T}{2}}\right) \tag{7.26}$$

These equations along with equations 7.19 will give the complete solution for the x and y motion. Now, we have to find S_{cl}. For that,

$$\frac{m}{2}\int_0^T \dot{x}^2 dt = \frac{m}{2}\left[(x\dot{x})_0^T - \int_0^T x\ddot{x} dt\right] \tag{7.27}$$
$$= \frac{m}{2}(x\dot{x})_0^T - \frac{eB}{2c}\int_0^T x\dot{y} dt \tag{7.28}$$

The above is obtained using equation 7.6. Similarly, we get two more results,

$$\frac{m}{2}\int_0^T \dot{y}^2 dt = \frac{m}{2}(y\dot{y})_0^T + \frac{eB}{2c}\int_0^T y\dot{x} dt \tag{7.29}$$
$$\frac{m}{2}\int_0^T \dot{z}^2 dt = \frac{m(z_b - z_a)^2}{2T} \tag{7.30}$$

Hence the expression for S_{cl} will become,

$$S_{cl} = \int_0^T \left[\frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + \frac{eB}{2c}(x\dot{y} - y\dot{x})\right] dt \tag{7.31}$$

$$= \frac{m}{2}\left[(x\dot{x})_0^T + (y\dot{y})_0^T\right] + \frac{m(z_b - z_a)^2}{2T} \tag{7.32}$$

Now note that,

$$\dot{x}(t) = \omega(B_1 \cos\omega t - A_1 \sin\omega t) = \omega(y(t) - C_2) \tag{7.33}$$

And similarly,

$$\dot{y}(t) = \omega(C_1 - x(t)) \tag{7.34}$$

Therefore we have,

$$S_{cl} = \frac{m(z_b - z_a)^2}{2T} + \frac{m\omega}{2}\left[x(y - C_2) + y(C_1 - x)\right]_0^T \tag{7.35}$$

Using the constants from equation 7.23,

$$S_{cl} = \frac{m(z_b - z_a)^2}{2T} + \frac{m\omega}{4}\left[(y_b - y_a)\left(x_a + x_b + \frac{y_b - y_a}{\tan\frac{\omega T}{2}}\right) - (x_b - x_a)\left(y_a + y_b + \frac{x_a - x_b}{\tan\frac{\omega T}{2}}\right)\right] \tag{7.36}$$

Further simplifications will yield,

$$S_{cl} = \frac{m(z_b - z_a)^2}{2T} + \frac{m\omega}{4\tan\frac{\omega T}{2}}\left((x_b - x_a)^2 + (y_b - y_a)^2\right) + \frac{m\omega}{2}(y_b x_a - x_b y_a) \tag{7.37}$$

Now using equation 4.7,

$$K(b,a) = F(T)exp\left\{\frac{im}{2\hbar}\left[\frac{(z_b - z_a)^2}{T} + \frac{\omega/2}{\tan\frac{\omega T}{2}}[(x_b - x_a)^2 + (y_b - y_a)^2] + \omega(y_b x_a - x_b y_a)\right]\right\} \tag{7.38}$$

Now we have to find $F(T)$ to get the complete solution for the problem. The prefactor for the z coordinate part is just the good old free particle prefactor $\sqrt{\frac{m}{2\pi i\hbar T}}$. The xy part is independent of the z coordinate change, which is evident from the equations of motion. To find the prefactor for xy part, let us take

$$F(T) = \sqrt{\frac{m}{2\pi i\hbar T}}G(T) \tag{7.39}$$

Hence $G(T)$ is the prefactor for xy part. Choose any (x_c, y_c, t_c) and let $t_b - t_c = t$ and $t_c - t_b = s$.

$$\begin{aligned} G(T) &= K(0,0,t_b;0,0,t_a) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} K(0,0,t_b;x_c,y_c,t_c)K(x_c,y_c,t_c;0,0,t_a)dx_c dy_c \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} G(t)exp\left[\frac{im\omega}{4\hbar\tan\frac{\omega t}{2}}(x_c^2 + y_c^2)\right]G(s)exp\left[\frac{im\omega}{4\hbar\tan\frac{\omega s}{2}}(x_c^2 + y_c^2)\right]dx_c dy_c \end{aligned}$$

Using equation 2.7 we get,

$$G(T) = G(t)G(s)\left[i\sqrt{\frac{\pi}{\frac{im\omega}{4\hbar}\left(\frac{1}{\tan\frac{\omega t}{2}} + \frac{1}{\tan\frac{\omega s}{2}}\right)}}\right]^2 = -G(t)G(s)\frac{4\pi\hbar\sin\frac{\omega t}{2}\sin\frac{\omega s}{2}}{im\omega\sin\frac{\omega T}{2}} \tag{7.40}$$

Take

$$G(t) = \frac{m\omega}{4\pi i\hbar\sin\frac{\omega t}{2}}g(t) \tag{7.41}$$

Substituting this in the above equation will yield,

$$g(T) = g(t)g(s) \tag{7.42}$$

Choose $g(t) = 1$ for all t. Therefore,

$$G(T) = \frac{m\omega}{4\pi i\hbar \sin\frac{\omega T}{2}} \tag{7.43}$$

Hence,

$$F(T) = \sqrt{\frac{m}{2\pi i\hbar T}\frac{m\omega}{4\pi i\hbar \sin\frac{\omega t}{2}}} = \left(\frac{m}{2\pi i\hbar T}\right)^{3/2} \frac{\omega T/2}{\sin\frac{\omega T}{2}} \tag{7.44}$$

Hence the Propagator for the Landau Problem is,

$$K(b,a) = \left(\frac{m}{2\pi i\hbar T}\right)^{3/2} \frac{\omega T/2}{\sin\frac{\omega T}{2}}\, exp\left\{\frac{im}{2\hbar}\left[\frac{(z_b - z_a)^2}{T} + \frac{\omega/2}{\tan\frac{\omega T}{2}}[(x_b - x_a)^2 + (y_b - y_a)^2] + \omega(y_b x_a - x_b y_a)\right]\right\} \tag{7.45}$$

I tried to find the energy eigenstates of this problem using the method used in the case of SHO. But it is very difficult to find using this method.